BENCHMARK OVERVIEW

GPT-OSS-120B VIS Benchmark

without aiDAPTIV vs with aiDAPTIV (Cold / Warm)

True GPT-OSS-120B NP1-32 serving data. Same visual logic as the 70B report, with Cold and repeated-prompt Warm separated.

1) Benchmark scope

Model	gpt-oss-120b
NP levels	1 / 2 / 4 / 8 / 16 / 32
Input tokens	1K / 2K / 4K / 8K / 16K
Output tokens	1K / 2K

2) Comparison modes

without aiDAPTIV	3-run baseline average (Run1/2/3)
with aiDAPTIV (Cold)	Run1
with aiDAPTIV (Warm)	Average of Run2 and Run3

3) Metrics in this report

TTFT (ms)	Time from request start to first token
TPOT (ms/token)	Median TPOT based decode timing
Aggregate throughput	Raw vLLM Output token throughput (tok/s)
Normalized throughput	Aggregate throughput / NP (tok/s/user)

4) Reading guide

TTFT / TPOT	Lower is better
Throughput	Higher is better
Warm state	Repeated same-prompt runs on the same live server
TPOT robustness	Median TPOT; invalid short-output warm runs filtered for TPOT only

Method note: Warm = repeated-prompt Run2/Run3 behavior. The runner remained live; this is not the formal random_run_phase=2 80% reuse mode.

Without aiDAPTIV — Cold (Run1) — Warm (Run2/Run3 avg)

TTFT

Time to first token (ms). Lower is better.

Linear X/Y in every NP panel; each panel uses its own Y max (shown) for readability.

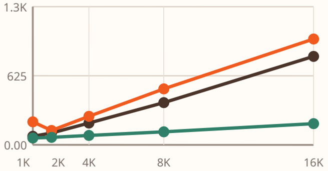
FORMULA **TTFT (ms) = mean(first-token time - request-start time)**

Reduction (%) = (Without - Warm) / Without x 100

Output 1K

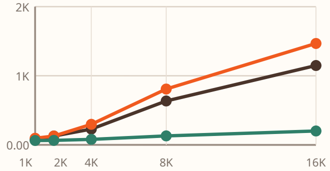
Linear numeric Input-token axis; 1K, 2K, 4K, 8K, 16K are plotted at proportional X positions.

NP1 (1 user)



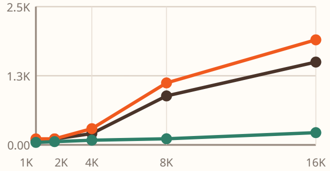
Input	Without	Cold	Warm	Reduction
1K	77	208	62	+20.2%
2K	107	130	68	+36.8%
4K	197	258	85	+57.0%
8K	381	506	118	+69.1%
16K	800	957	191	+76.1%

NP2 (2 users)



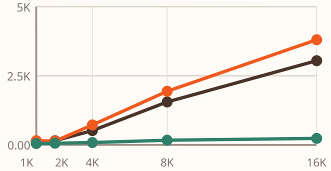
Input	Without	Cold	Warm	Reduction
1K	73	95	63	+14.0%
2K	127	128	64	+49.5%
4K	227	296	79	+65.1%
8K	635	807	129	+79.7%
16K	1,147	1,467	200	+82.6%

NP4 (4 users)



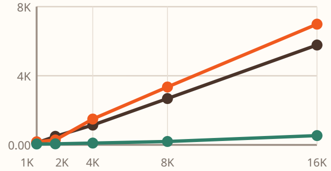
Input	Without	Cold	Warm	Reduction
1K	80	108	44	+44.5%
2K	105	108	58	+44.4%
4K	204	292	84	+58.7%
8K	884	1,121	110	+87.6%
16K	1,497	1,899	221	+85.2%

NP8 (8 users)



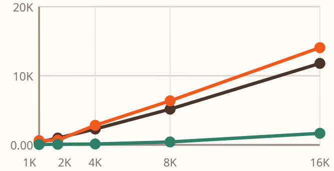
Input	Without	Cold	Warm	Reduction
1K	108	152	49	+55.1%
2K	152	134	59	+61.0%
4K	514	720	84	+83.7%
8K	1,552	1,937	167	+89.2%
16K	3,043	3,800	234	+92.3%

NP16 (16 users)



Input	Without	Cold	Warm	Reduction
1K	114	177	50	+56.0%
2K	492	282	57	+88.5%
4K	1,141	1,495	101	+91.1%
8K	2,678	3,347	193	+92.8%
16K	5,775	6,983	535	+90.7%

NP32 (32 users)

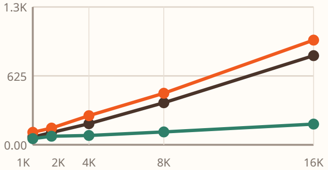


Input	Without	Cold	Warm	Reduction
1K	382	614	52	+86.4%
2K	980	681	85	+91.3%
4K	2,285	2,828	129	+94.4%
8K	5,165	6,369	424	+91.8%
16K	11,787	14,058	1,661	+85.9%

Output 2K

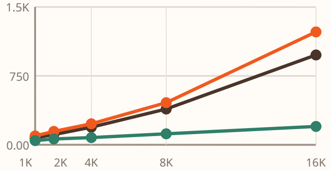
Linear numeric Input-token axis; 1K, 2K, 4K, 8K, 16K are plotted at proportional X positions.

NP1 (1 user)



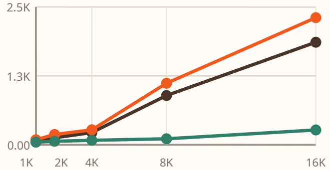
Input	Without	Cold	Warm	Reduction
1K	70	113	56	+20.8%
2K	111	151	77	+30.6%
4K	191	264	84	+56.0%
8K	381	467	117	+69.4%
16K	808	949	188	+76.8%

NP2 (2 users)



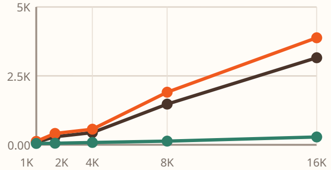
Input	Without	Cold	Warm	Reduction
1K	74	96	46	+38.4%
2K	111	145	63	+43.1%
4K	192	229	78	+59.2%
8K	389	458	119	+69.4%
16K	977	1,228	199	+79.6%

NP4 (4 users)



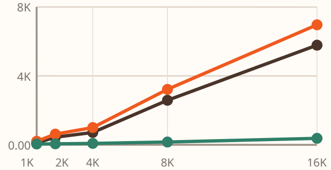
Input	Without	Cold	Warm	Reduction
1K	72	91	47	+34.3%
2K	126	186	63	+50.3%
4K	222	271	81	+63.8%
8K	893	1,116	108	+87.9%
16K	1,860	2,305	268	+85.6%

NP8 (8 users)



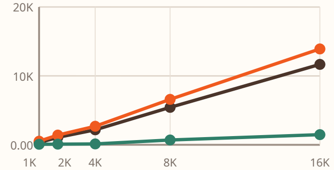
Input	Without	Cold	Warm	Reduction
1K	89	123	48	+46.3%
2K	291	411	56	+80.9%
4K	440	563	82	+81.4%
8K	1,475	1,907	131	+91.1%
16K	3,155	3,882	282	+91.1%

NP16 (16 users)



Input	Without	Cold	Warm	Reduction
1K	108	204	45	+58.1%
2K	437	617	55	+87.3%
4K	713	1,000	79	+88.9%
8K	2,582	3,216	158	+93.9%
16K	5,781	6,961	375	+93.5%

NP32 (32 users)



Input	Without	Cold	Warm	Reduction
1K	309	518	67	+78.2%
2K	1,080	1,405	89	+91.8%
4K	2,174	2,684	128	+94.1%
8K	5,429	6,583	696	+87.2%
16K	11,664	13,898	1,471	+87.4%

Without aiDAPTIV — Cold (Run1) — Warm (Run2/Run3 avg)

TPOT

Median time per output token (ms/token). Lower is better.

Linear X/Y in every NP panel; each panel uses its own Y max (shown) for readability.

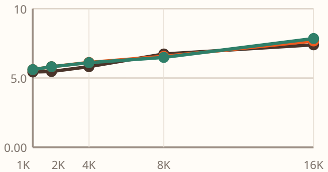
FORMULA **TPOT = (E2E latency - TTFT) / (output tokens - 1)**

Reduction (%) = (Without - Warm) / Without x 100

Output 1K

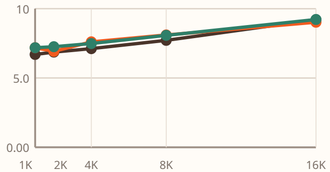
Linear numeric Input-token axis; 1K, 2K, 4K, 8K, 16K are plotted at proportional X positions.

NP1 (1 user)



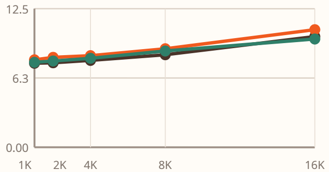
Input	Without	Cold	Warm	Reduction
1K	5.43	5.60	5.60	-3.1%
2K	5.47	5.81	5.81	-6.3%
4K	5.82	6.12	6.11	-5.0%
8K	6.73	6.56	6.48	+3.7%
16K	7.39	7.67	7.84	-6.2%

NP2 (2 users)



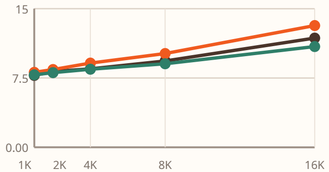
Input	Without	Cold	Warm	Reduction
1K	6.69	7.18	7.18	-7.3%
2K	6.86	6.92	7.25	-5.7%
4K	7.12	7.60	7.48	-5.1%
8K	7.71	8.10	8.07	-4.6%
16K	9.21	9.02	9.21	-0.1%

NP4 (4 users)



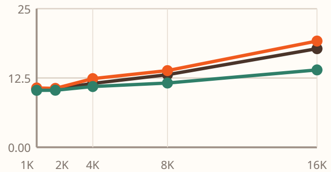
Input	Without	Cold	Warm	Reduction
1K	7.57	7.88	7.66	-1.1%
2K	7.61	8.11	7.78	-2.2%
4K	7.83	8.26	8.02	-2.4%
8K	8.34	8.88	8.68	-4.1%
16K	9.98	10.60	9.77	+2.2%

NP8 (8 users)



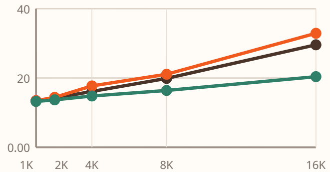
Input	Without	Cold	Warm	Reduction
1K	7.77	8.11	7.83	-0.8%
2K	8.20	8.41	8.07	+1.6%
4K	8.48	9.11	8.44	+0.5%
8K	9.34	10.14	9.03	+3.3%
16K	11.81	13.17	10.89	+7.7%

NP16 (16 users)



Input	Without	Cold	Warm	Reduction
1K	10.49	10.69	10.27	+2.1%
2K	10.52	10.61	10.29	+2.2%
4K	11.51	12.38	10.96	+4.8%
8K	13.08	13.82	11.59	+11.4%
16K	17.77	19.16	13.95	+21.5%

NP32 (32 users)

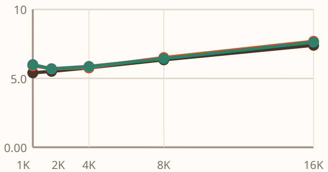


Input	Without	Cold	Warm	Reduction
1K	13.43	13.43	13.22	+1.6%
2K	14.20	14.40	13.69	+3.6%
4K	16.11	17.71	14.80	+8.1%
8K	19.86	21.06	16.38	+17.5%
16K	29.55	32.87	20.38	+31.0%

Output 2K

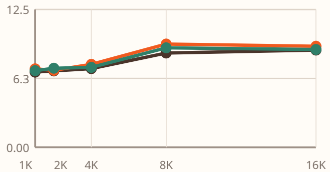
Linear numeric Input-token axis; 1K, 2K, 4K, 8K, 16K are plotted at proportional X positions.

NP1 (1 user)



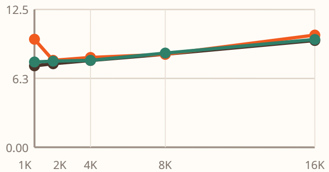
Input	Without	Cold	Warm	Reduction
1K	5.40	5.93	5.99	-11.0%
2K	5.51	5.69	5.70	-3.4%
4K	5.76	5.80	5.87	-1.8%
8K	6.35	6.51	6.44	-1.4%
16K	7.40	7.69	7.63	-3.1%

NP2 (2 users)



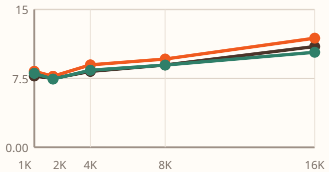
Input	Without	Cold	Warm	Reduction
1K	6.81	7.11	6.95	-1.9%
2K	6.92	6.98	7.17	-3.5%
4K	7.13	7.52	7.25	-1.6%
8K	8.54	9.36	9.02	-5.5%
16K	8.81	9.16	8.87	-0.7%

NP4 (4 users)



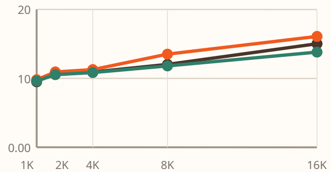
Input	Without	Cold	Warm	Reduction
1K	7.39	9.80	7.73	-4.6%
2K	7.57	7.88	7.81	-3.1%
4K	7.87	8.14	7.88	-0.2%
8K	8.44	8.43	8.54	-1.2%
16K	9.68	10.17	9.77	-1.0%

NP8 (8 users)



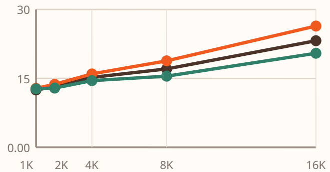
Input	Without	Cold	Warm	Reduction
1K	7.75	8.27	8.04	-3.8%
2K	7.49	7.73	7.42	+0.9%
4K	8.26	8.97	8.39	-1.6%
8K	8.95	9.60	8.94	+0.1%
16K	10.96	11.87	10.34	+5.7%

NP16 (16 users)



Input	Without	Cold	Warm	Reduction
1K	9.49	9.80	9.61	-1.2%
2K	10.82	10.93	10.53	+2.7%
4K	10.91	11.27	10.84	+0.6%
8K	12.02	13.51	11.80	+1.9%
16K	15.02	16.08	13.80	+8.1%

NP32 (32 users)



Input	Without	Cold	Warm	Reduction
1K	12.47	12.81	12.71	-2.0%
2K	13.18	13.71	12.88	+2.2%
4K	15.21	15.96	14.50	+4.6%
8K	17.05	18.80	15.47	+9.3%
16K	23.19	26.39	20.46	+11.8%

THROUGHPUT

Throughput (Aggregate)

Raw vLLM output token throughput (tok/s). Higher is better.

Linear X/Y in every NP panel; each panel uses its own Y max (shown) for readability.

Without aiDAPTIV — Cold (Run1) — Warm (Run2/Run3 avg)

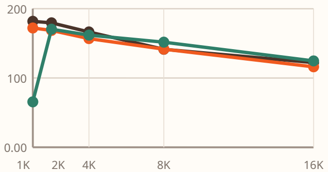
FORMULA **Aggregate tok/s = total generated output tokens / benchmark duration**

Gain (%) = (Warm / Without - 1) x 100

Output 1K

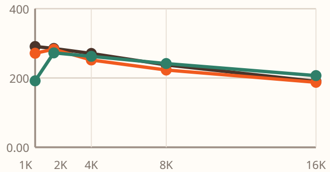
Linear numeric Input-token axis; 1K, 2K, 4K, 8K, 16K are plotted at proportional X positions.

NP1 (1 user)



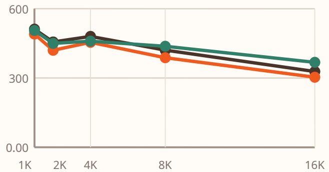
Input	Without	Cold	Warm	Gain
1K	181.7	172.1	65.6	-63.9%
2K	179.5	168.5	170.3	-5.1%
4K	166.3	156.8	161.8	-2.7%
8K	141.3	141.6	151.6	+7.3%
16K	122.2	116.0	124.6	+2.0%

NP2 (2 users)



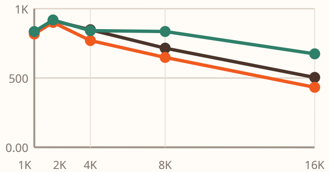
Input	Without	Cold	Warm	Gain
1K	290.8	271.4	191.6	-34.1%
2K	285.1	282.9	272.3	-4.5%
4K	270.6	251.9	262.3	-3.0%
8K	237.6	222.9	241.6	+1.7%
16K	190.1	187.6	206.7	+8.7%

NP4 (4 users)



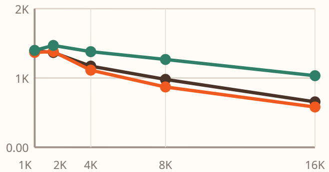
Input	Without	Cold	Warm	Gain
1K	511.1	489.8	505.3	-1.2%
2K	455.7	419.8	449.6	-1.3%
4K	479.7	454.2	459.0	-4.3%
8K	420.4	387.4	437.1	+4.0%
16K	327.9	303.4	367.5	+12.1%

NP8 (8 users)



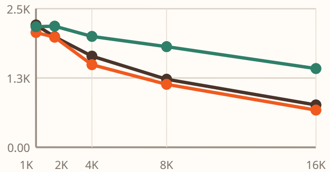
Input	Without	Cold	Warm	Gain
1K	833.7	817.2	833.8	+0.0%
2K	912.6	900.9	918.1	+0.6%
4K	847.5	770.0	841.3	-0.7%
8K	715.0	648.1	835.1	+16.8%
16K	503.7	432.5	673.8	+33.8%

NP16 (16 users)



Input	Without	Cold	Warm	Gain
1K	1393.3	1368.1	1397.9	+0.3%
2K	1366.9	1380.1	1468.8	+7.5%
4K	1170.9	1112.0	1378.9	+17.8%
8K	979.1	869.8	1266.9	+29.4%
16K	655.2	581.1	1032.9	+57.6%

NP32 (32 users)

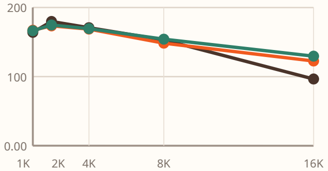


Input	Without	Cold	Warm	Gain
1K	2205.9	2069.8	2174.0	-1.4%
2K	1991.6	1985.5	2183.2	+9.6%
4K	1644.2	1488.9	1999.1	+21.6%
8K	1226.5	1134.4	1814.0	+47.9%
16K	763.7	669.6	1418.8	+85.8%

Output 2K

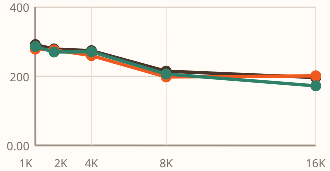
Linear numeric Input-token axis; 1K, 2K, 4K, 8K, 16K are plotted at proportional X positions.

NP1 (1 user)



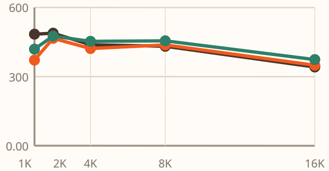
Input	Without	Cold	Warm	Gain
1K	163.9	167.1	166.4	+1.5%
2K	179.7	173.4	174.5	-2.9%
4K	170.7	168.7	169.4	-0.8%
8K	153.0	148.3	154.1	+0.7%
16K	96.5	122.5	129.6	+34.3%

NP2 (2 users)



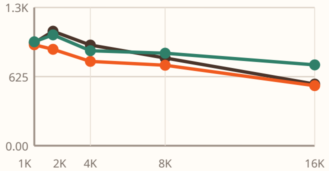
Input	Without	Cold	Warm	Gain
1K	291.6	278.9	286.4	-1.8%
2K	279.2	276.4	270.4	-3.2%
4K	274.2	260.0	271.0	-1.2%
8K	214.9	198.2	206.8	-3.8%
16K	196.6	201.2	172.4	-12.3%

NP4 (4 users)



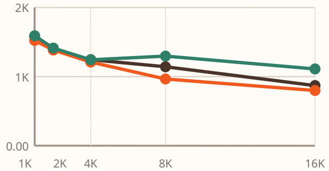
Input	Without	Cold	Warm	Gain
1K	484.1	371.4	419.2	-13.4%
2K	488.4	465.9	475.7	-2.6%
4K	438.2	421.6	452.9	+3.4%
8K	431.4	437.5	455.3	+5.5%
16K	341.7	349.3	373.7	+9.4%

NP8 (8 users)



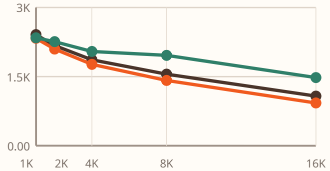
Input	Without	Cold	Warm	Gain
1K	929.5	912.9	939.3	+1.1%
2K	1034.9	871.8	1002.5	-3.1%
4K	909.6	762.5	858.9	-5.6%
8K	792.8	727.5	836.0	+5.5%
16K	557.0	542.7	730.8	+31.2%

NP16 (16 users)



Input	Without	Cold	Warm	Gain
1K	1577.0	1522.7	1590.9	+0.9%
2K	1391.3	1383.8	1412.4	+1.5%
4K	1244.9	1210.3	1243.0	-0.2%
8K	1142.6	965.4	1296.7	+13.5%
16K	869.3	799.0	1110.8	+27.8%

NP32 (32 users)



Input	Without	Cold	Warm	Gain
1K	2410.1	2330.7	2342.1	-2.8%
2K	2173.9	2097.3	2257.5	+3.8%
4K	1863.2	1764.9	2045.0	+9.8%
8K	1552.4	1416.0	1961.6	+26.4%
16K	1076.2	927.2	1478.0	+37.3%

THROUGHPUT

Throughput (Normalized)

Aggregate throughput divided by NP (tok/s/user). Higher is better.

Linear X/Y in every NP panel; each panel uses its own Y max (shown) for readability.

Without aiDAPTIV | Cold (Run1) | Warm (Run2/Run3 avg)

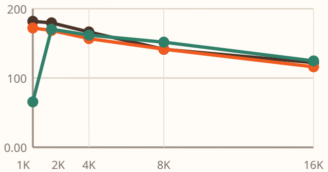
FORMULA **Normalized tok/s/user = Aggregate tok/s / NP**

Gain (%) = (Warm / Without - 1) x 100

Output 1K

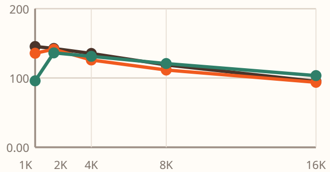
Linear numeric Input-token axis; 1K, 2K, 4K, 8K, 16K are plotted at proportional X positions.

NP1 (1 user)



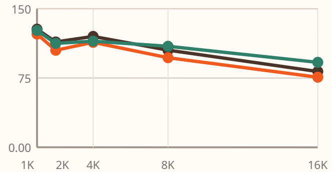
Input	Without	Cold	Warm	Gain
1K	181.69	172.11	65.59	-63.9%
2K	179.54	168.52	170.32	-5.1%
4K	166.32	156.78	161.75	-2.7%
8K	141.32	141.60	151.60	+7.3%
16K	122.16	116.05	124.60	+2.0%

NP2 (2 users)



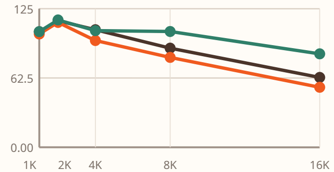
Input	Without	Cold	Warm	Gain
1K	145.41	135.69	95.80	-34.1%
2K	142.54	141.45	136.14	-4.5%
4K	135.28	125.94	131.16	-3.0%
8K	118.80	111.45	120.78	+1.7%
16K	95.03	93.81	103.34	+8.7%

NP4 (4 users)



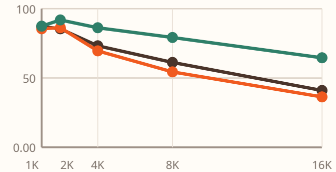
Input	Without	Cold	Warm	Gain
1K	127.79	122.45	126.31	-1.2%
2K	113.92	104.96	112.41	-1.3%
4K	119.91	113.55	114.75	-4.3%
8K	105.11	96.86	109.27	+4.0%
16K	81.97	75.84	91.89	+12.1%

NP8 (8 users)



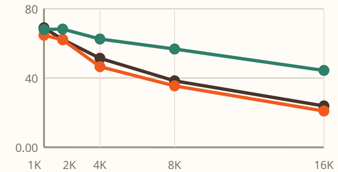
Input	Without	Cold	Warm	Gain
1K	104.21	102.15	104.22	+0.0%
2K	114.08	112.61	114.77	+0.6%
4K	105.93	96.25	105.16	-0.7%
8K	89.37	81.02	104.39	+16.8%
16K	62.96	54.06	84.22	+33.8%

NP16 (16 users)



Input	Without	Cold	Warm	Gain
1K	87.08	85.51	87.37	+0.3%
2K	85.43	86.26	91.80	+7.5%
4K	73.18	69.50	86.18	+17.8%
8K	61.19	54.36	79.18	+29.4%
16K	40.95	36.32	64.55	+57.6%

NP32 (32 users)

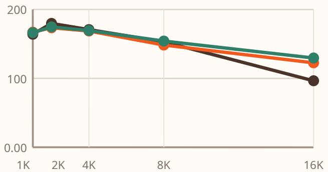


Input	Without	Cold	Warm	Gain
1K	68.93	64.68	67.94	-1.4%
2K	62.24	62.05	68.23	+9.6%
4K	51.38	46.53	62.47	+21.6%
8K	38.33	35.45	56.69	+47.9%
16K	23.86	20.93	44.34	+85.8%

Output 2K

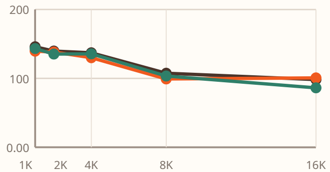
Linear numeric Input-token axis; 1K, 2K, 4K, 8K, 16K are plotted at proportional X positions.

NP1 (1 user)



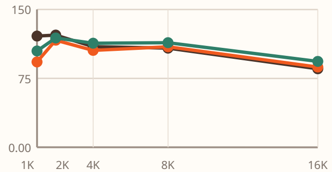
Input	Without	Cold	Warm	Gain
1K	163.95	167.08	166.41	+1.5%
2K	179.70	173.39	174.53	-2.9%
4K	170.74	168.71	169.38	-0.8%
8K	153.04	148.29	154.08	+0.7%
16K	96.47	122.48	129.55	+34.3%

NP2 (2 users)



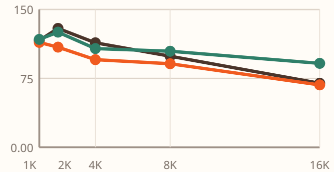
Input	Without	Cold	Warm	Gain
1K	145.81	139.44	143.22	-1.8%
2K	139.59	138.21	135.19	-3.2%
4K	137.08	129.99	135.49	-1.2%
8K	107.43	99.09	103.39	-3.8%
16K	98.29	100.61	86.22	-12.3%

NP4 (4 users)



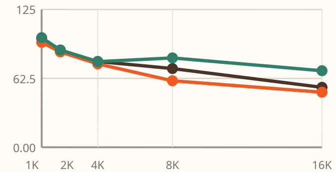
Input	Without	Cold	Warm	Gain
1K	121.04	92.84	104.79	-13.4%
2K	122.10	116.47	118.92	-2.6%
4K	109.54	105.39	113.22	+3.4%
8K	107.85	109.37	113.83	+5.5%
16K	85.43	87.33	93.43	+9.4%

NP8 (8 users)



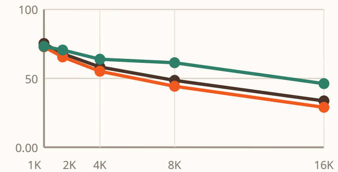
Input	Without	Cold	Warm	Gain
1K	116.19	114.11	117.42	+1.1%
2K	129.36	108.97	125.31	-3.1%
4K	113.71	95.32	107.36	-5.6%
8K	99.10	90.94	104.50	+5.5%
16K	69.62	67.84	91.35	+31.2%

NP16 (16 users)



Input	Without	Cold	Warm	Gain
1K	98.56	95.17	99.43	+0.9%
2K	86.96	86.49	88.28	+1.5%
4K	77.80	75.64	77.69	-0.2%
8K	71.41	60.34	81.05	+13.5%
16K	54.33	49.94	69.43	+27.8%

NP32 (32 users)



Input	Without	Cold	Warm	Gain
1K	75.32	72.83	73.19	-2.8%
2K	67.94	65.54	70.55	+3.8%
4K	58.23	55.15	63.91	+9.8%
8K	48.51	44.25	61.30	+26.4%
16K	33.63	28.98	46.19	+37.3%

SUMMARY

aiDAPTIV impact on latency, decode efficiency, and throughput

GPT-OSS-120B VIS benchmark, NP1-32; summary emphasizes the 16K long-context boundary at NP16 and NP32.

TTFT

Time to first token (ms)

Warm aiDAPTIV vs without aiDAPTIV

NP16 | Out1K
5,775 -> 535
+90.7% lower

NP16 | Out2K
5,781 -> 375
+93.5% lower

NP32 | Out1K
11,787 -> 1,661
+85.9% lower

NP32 | Out2K
11,664 -> 1,471
+87.4% lower

TPOT

Time per output token (ms/token)

Warm aiDAPTIV vs without aiDAPTIV

NP16 | Out1K
17.77 -> 13.95
+21.5% lower

NP16 | Out2K
15.02 -> 13.80
+8.1% lower

NP32 | Out1K
29.55 -> 20.38
+31.0% lower

NP32 | Out2K
23.19 -> 20.46
+11.8% lower

Aggregate throughput

Total server output capacity (tok/s)

Warm aiDAPTIV vs without aiDAPTIV

NP16 | Out1K
655.2 -> 1032.9
+57.6% higher

NP16 | Out2K
869.3 -> 1110.8
+27.8% higher

NP32 | Out1K
763.7 -> 1418.8
+85.8% higher

NP32 | Out2K
1076.2 -> 1478.0
+37.3% higher

Normalized throughput

Per-user serving capacity (tok/s/user)

Warm aiDAPTIV vs without aiDAPTIV

NP16 | Out1K
40.95 -> 64.55
+57.6% higher

NP16 | Out2K
54.33 -> 69.43
+27.8% higher

NP32 | Out1K
23.86 -> 44.34
+85.8% higher

NP32 | Out2K
33.63 -> 46.19
+37.3% higher

Key interpretation

- Warm aiDAPTIV sharply reduces TTFT at long context: NP16/16K Out2K -93.5%; NP32/16K Out2K -87.4%.
- Decode efficiency improves more moderately than TTFT: NP32/16K TPOT falls by 31.0% (Out1K) and 11.8% (Out2K).
- Aggregate and normalized throughput gains grow with concurrency and long context; NP32/16K reaches +85.8% (Out1K) and +37.3% (Out2K).
- Cold Run1 can be slower than baseline; the strongest benefit appears after warm repeated-prompt reuse.